

Jaume **Ivars Grimalt**

ML & Computer Vision

Currently in Valencia, Spain · Open to relocation to Zurich, Munich, London

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Profile

I train detection models from scratch on proprietary datasets, scale training on HPC, and deploy them to production. At Mycrospace, I turned a Master's thesis into a computer vision product tested in five laboratories.

Technical Skills

AI & ML	Object detection, instance segmentation, density estimation, classification, CNNs, ViTs, foundational models, Neural ODEs, PINNs
Training	Python, PyTorch (DDP, AMP, torch.compile), Hugging Face, timm, multi-GPU HPC (SLURM)
Production	ONNX, Docker, GCP (Cloud Run, Cloud SQL, GCS), BullMQ/Redis, CI/CD
Product	NestJS, Prisma, PostgreSQL, MongoDB, Next.js, TypeScript, React Native/Expo, Stripe

Experience

Mycrospace — *Valencia, Spain*

Co-Founder & CTO — *Jan 2024 -- Oct 2026*

- CV for microbiology. Master's thesis (2024) → company (Jan 2026); tested in **5 labs**. 1st Prize IdeasUPV (2024).
- **Model.** Own ConvNeXt detector (YOLO/DQ-DETR-inspired), from scratch; 10--2000 colonies/image. **67 mAP50-95**; 92% recall / 93% precision.
- **Training. 12,500+** GPU-hours on Leonardo (two EuroHPC AI Factory allocations). **4x A100**, DDP/AMP/torch.compile/SLURM, checkpoint-resumed jobs. Own framework: k-fold CV, WBF ensembles.
- **Data. 1,500/5,000** images annotated with microbiologists. Weakly-supervised labelling (click → crop → 4-ch CNN heatmap → bbox); versioning + k-fold label scoring.
- **Product/infra.** BullMQ/Redis inference, signed-URL uploads, SSE/WebSocket; GCP (Cloud Run, Cloud SQL, GCS). NestJS/Prisma/PostgreSQL, MongoDB, Next.js, React Native/Expo, Stripe.

MIALAB -- Medical Imaging Analysis Laboratory, UPV — *Valencia, Spain*

Research Engineer (full-time) — *May 2025 -- Jan 2026*

- Volumetric brain MRI segmentation with nnU-Net; multi-scale feature strategies. Profiled volBrain (**700k+** volumes) from NIfTI ingest to voxel prediction for inference cost.

Tapstar — *Valencia, Spain*

Full-Stack Engineer (freelance) — *Jan 2025 -- May 2025*

- QR generation, back office and frontend for QR cards/stands; scan analytics and per-employee ratings.

Vesta-Z — *Valencia, Spain*

Data Engineer — *Jul 2023 -- Feb 2025*

- Full-stack cloud features; custom schema for cross-matching items across providers; LLM customer-support bot.

Neurocatching — *Valencia, Spain*

Data Scientist Intern — *Feb 2023 -- Aug 2023*

- Eye-gaze models for neuromarketing; data analysis for customer behaviour and sales.

Sciling — *Valencia, Spain*

Backend Developer Intern — *Jun 2022 -- Sep 2022*

- Conversational systems on GPT-3-era LLMs, applied NLP, and early diffusion models.

Education

M.Sc. Artificial Intelligence, Pattern Recognition & Digital Imaging — *Valencia, Spain*

UPV · GPA 8.7/10 · Thesis Honours (became Mycrospace) — *2023 -- 2024*

Technology Management — *Valencia / Munich*

CDTM (TUM/LMU) · 16 selected — *2024*

B.Sc. Computer Engineering — *Valencia, Spain*

UPV · GPA 8.3/10 · Honours in ML and Statistics — *2019 -- 2023*

Publications

- *Auto-N2N: Self-supervised MRI Denoising using Synthetic Pair Generation and Mixture of Experts.* S. Morell-Ortega et al., incl. **J. Ivars Grimalt**. Preprint, 2026. OpenReview

- *Detecting Cytopathic Effect in Cell Culture with Computer Vision.* W. Ragazzo, **J. Ivars Grimalt**. Under review.

- *Representational Dynamics in Residual Vision Networks* --- Independent Research. jaumeivars.com/research/representation-dynamics. Investigated continuous-depth dynamics in ConvNeXt, Swin and ResNet, evaluating finer numerical integration without retraining and comparing piecewise-constant versus interpolated residual blocks. Trained a weight-shared ConvNeXt-T stage from scratch on ImageNet-1k, achieving **80.34%** top-1 accuracy versus **80.96%** for the baseline with approximately **9.6M** fewer parameters. Analysed internal trajectory geometry, activation anisotropy and per-block weight corrections; a fine-tuned shared-block model reached **81.25%** top-1 accuracy.

Languages: Spanish (native), Catalan (native), English (C1)